

Rand

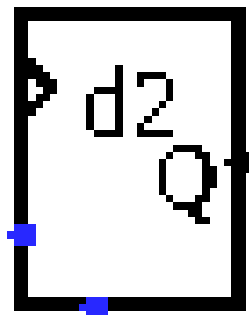
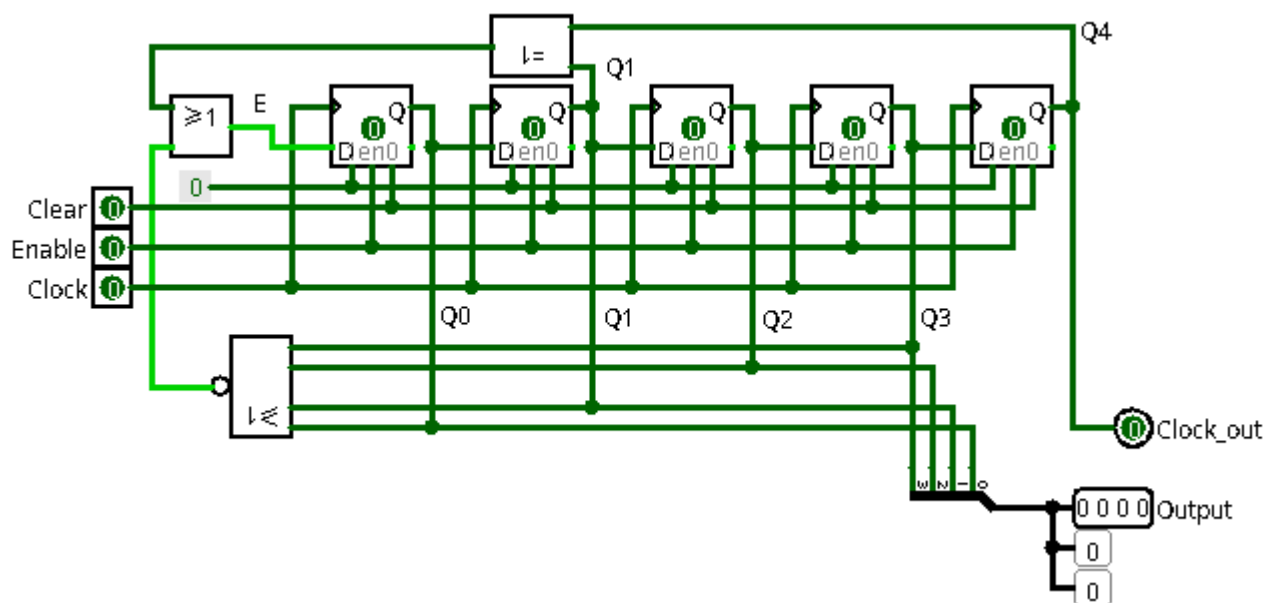
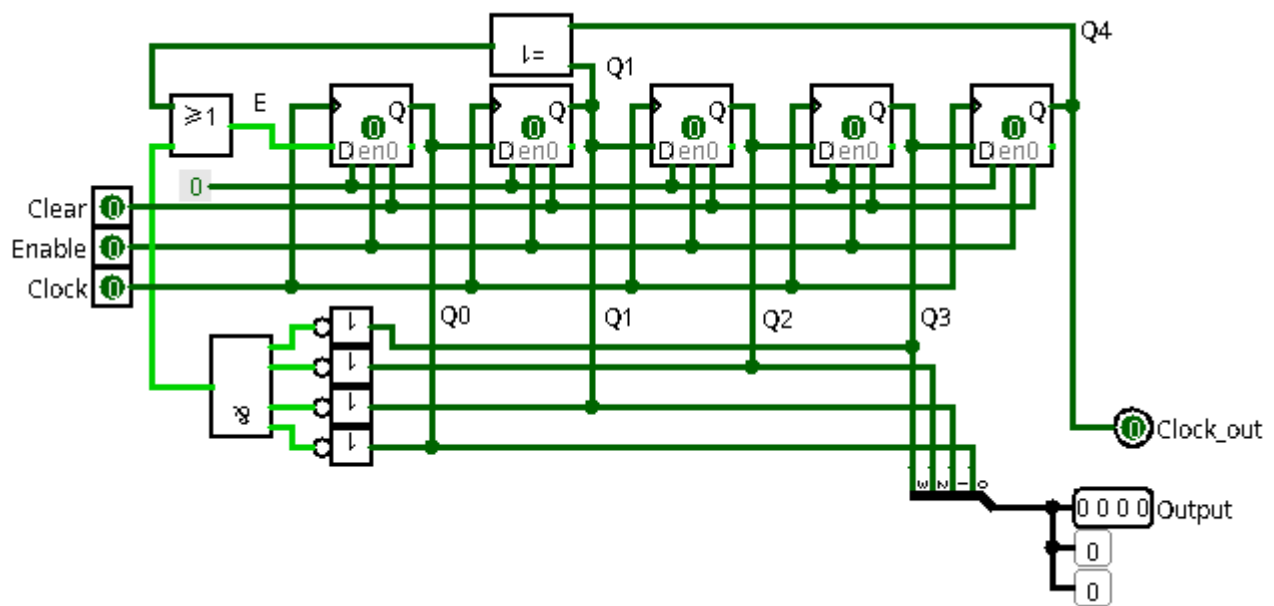


Table des matières

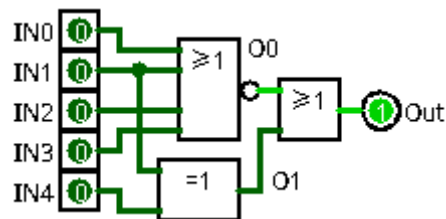
Rand.....	1
1) rand (semi-aléatoire): (SRAND4 - SRAND4M).....	3
2) Coupé en 2 parties :.....	4
2-1) Changement d'état de l'entrée : (CSTATE).....	4
2-2) Registre à décalage série : (SISO - SISO2).....	5
2-3) En totalité : (SRAND4S).....	7
3) Avec valeur maximale :.....	8
3-1) C'est quoi un masque : (MASK).....	8
3-2) Création d'un masque pour le nombre aléatoire :.....	8
3-3) Valeur max : (RVM4B).....	9
4) Avec valeur minimale :.....	10
5) 8 bits :.....	10

1) rand (semi-aléatoire): (SRAND4 - SRAND4M)



2) Coupé en 2 parties :

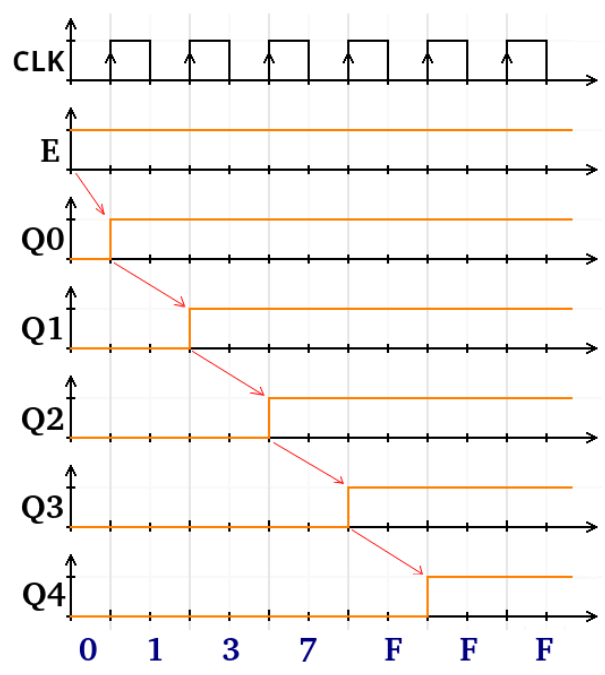
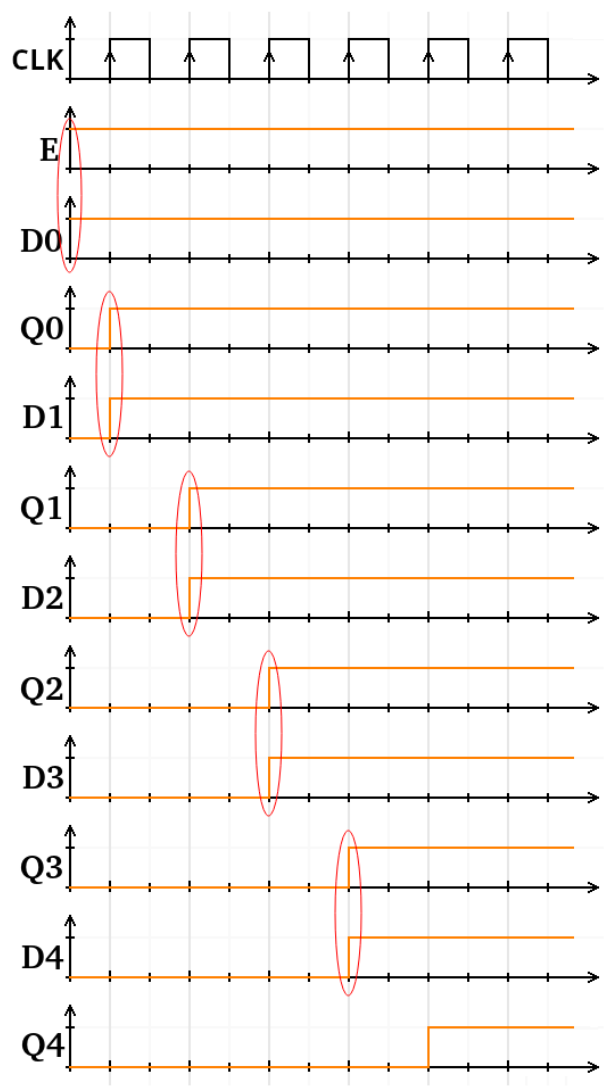
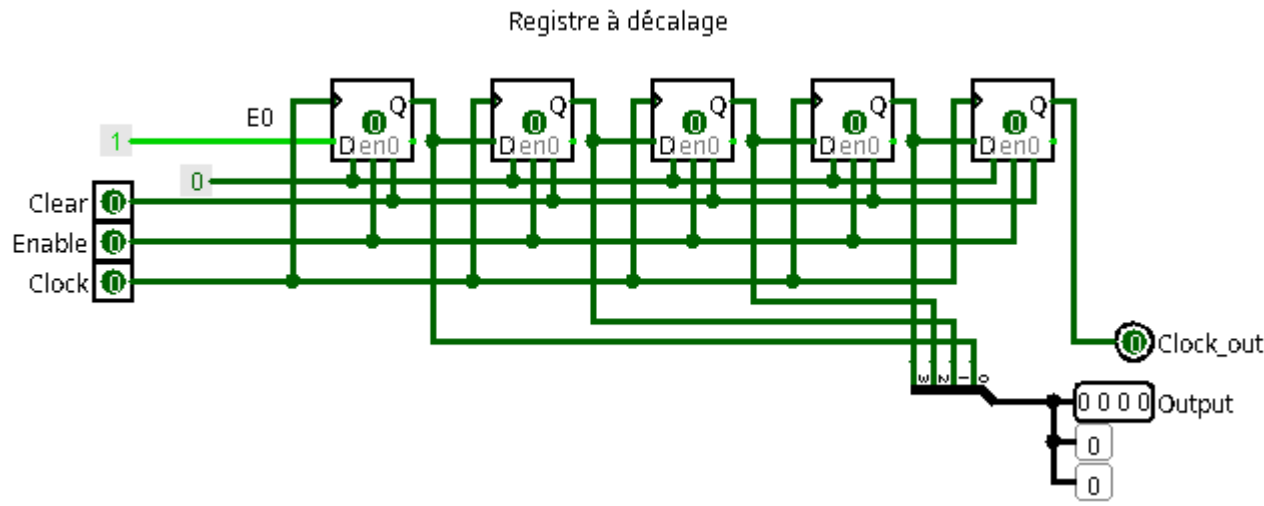
2-1) Changement d'état de l'entrée : (CSTATE)

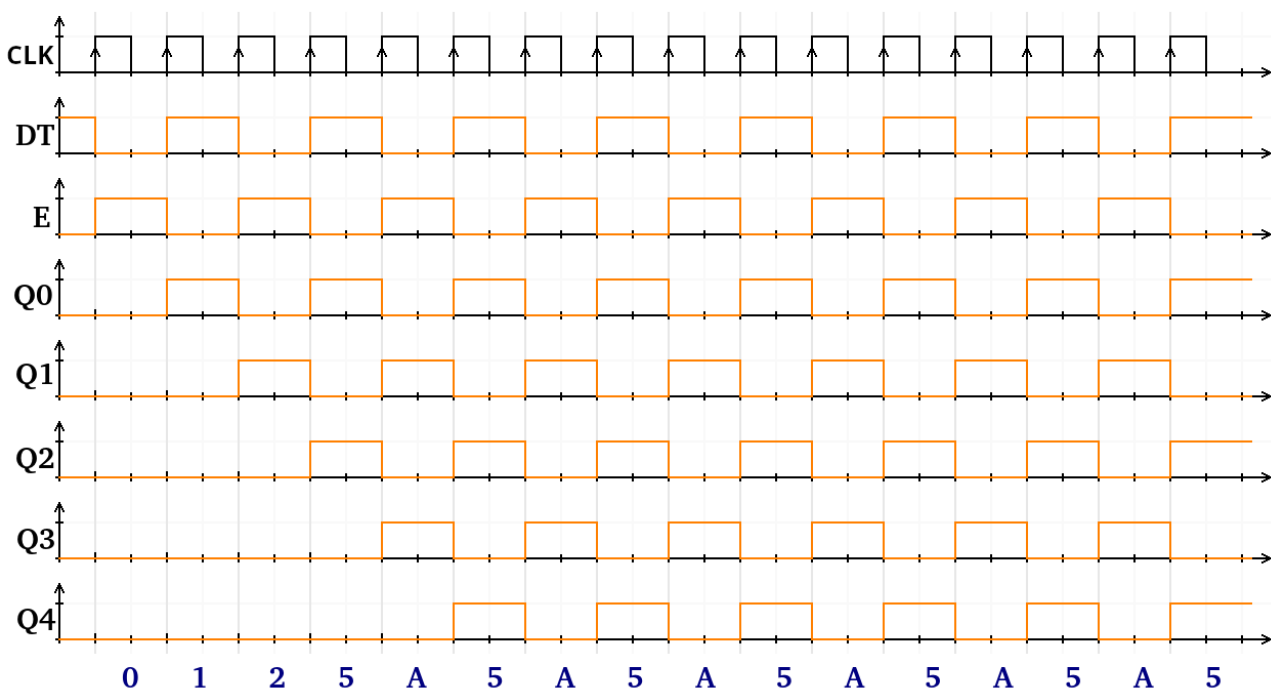


IN4	IN3	IN2	IN1	IN0	O0	O1	OUT
0	0	0	0	0	1	0	1
0	0	0	0	1	0	0	0
0	0	0	1	0	0	1	1
0	0	0	1	1	0	1	1
0	0	1	0	0	0	0	0
0	0	1	0	1	0	0	0
0	0	1	1	0	0	1	1
0	0	1	1	1	0	1	1
0	1	0	0	0	0	0	0
0	1	0	0	1	0	0	0
0	1	0	1	0	0	1	1
0	1	0	1	1	0	1	1
0	1	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	1	1	0	0	1	1
0	1	1	1	1	0	1	1
1	0	0	0	0	0	1	1
1	0	0	0	1	0	1	1
1	0	0	1	0	0	0	0
1	0	0	1	1	0	0	0
1	0	1	0	0	0	1	1
1	0	1	0	1	0	1	1
1	0	1	1	0	0	0	0
1	0	1	1	1	0	0	0
1	1	0	0	0	0	1	1
1	1	0	0	1	0	1	1
1	1	0	1	0	0	0	0
1	1	0	1	1	0	0	0
1	1	1	0	0	0	1	1
1	1	1	0	1	0	1	1
1	1	1	1	0	0	0	0
1	1	1	1	1	0	0	0

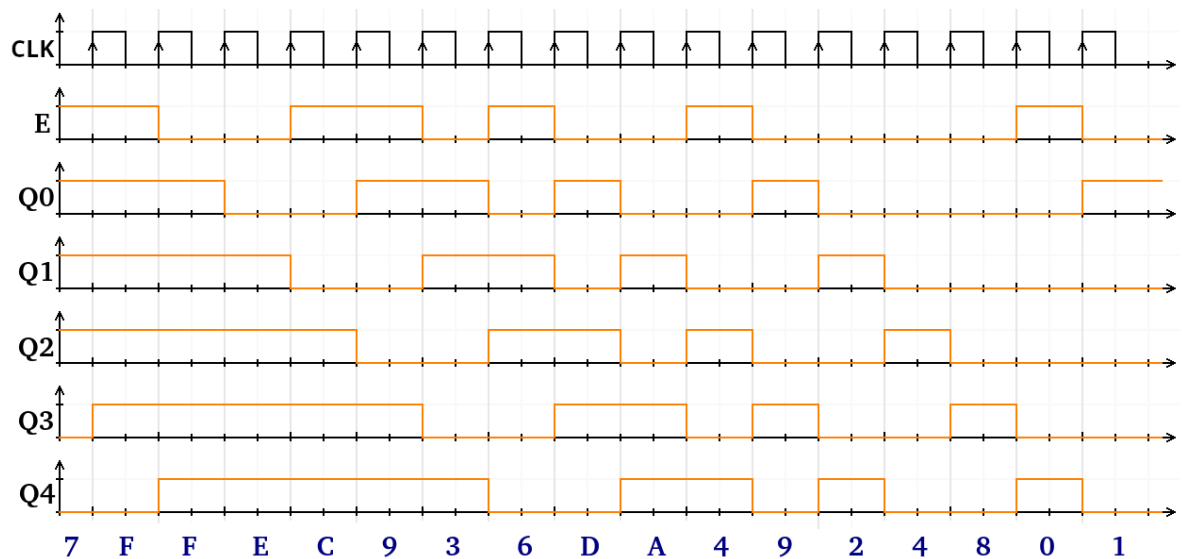
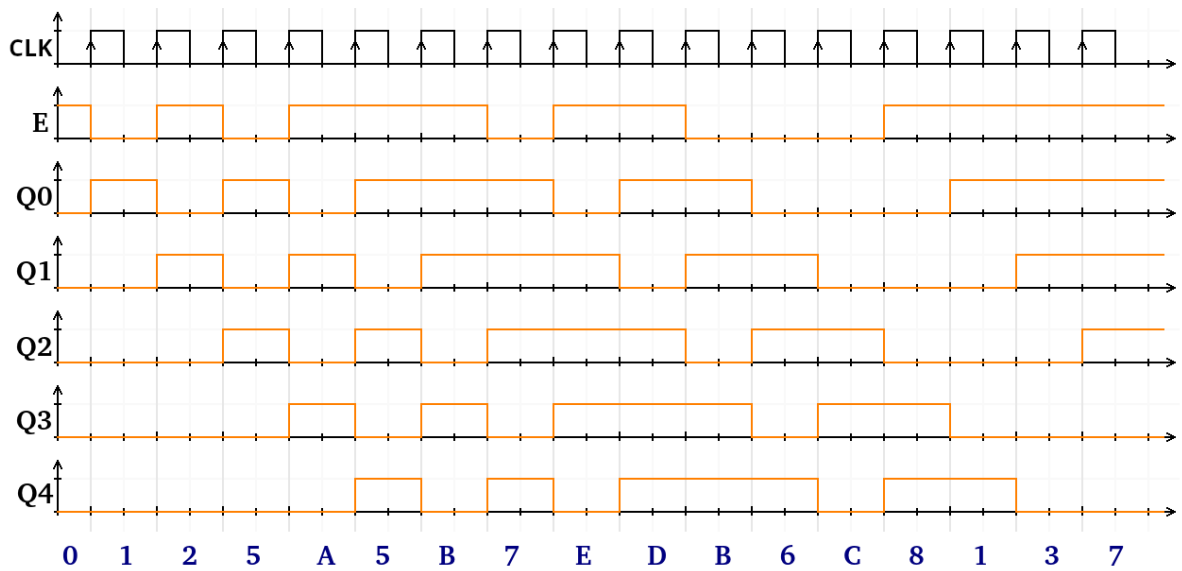
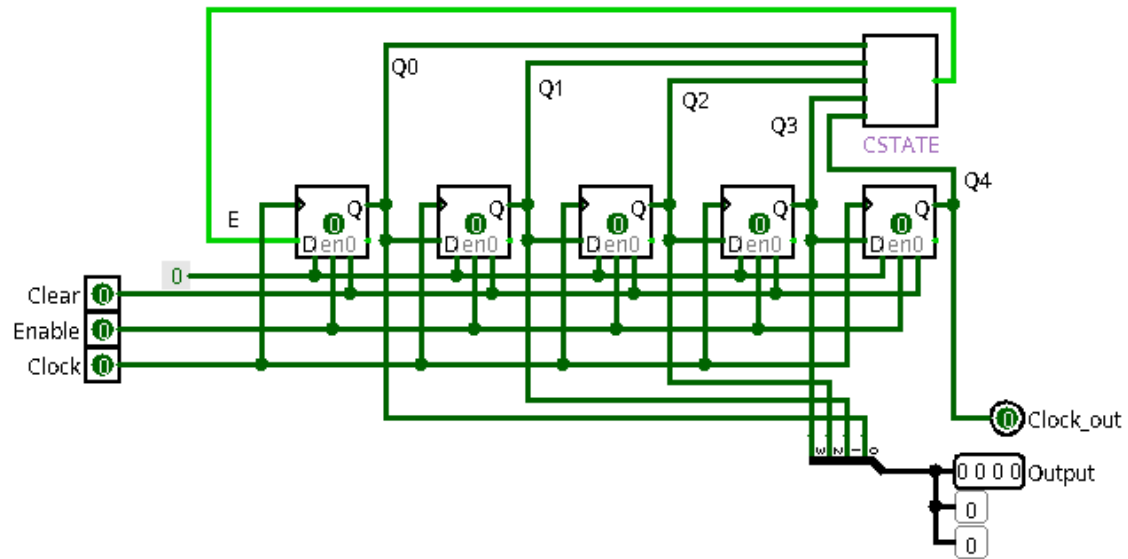
IN4	IN1	O1
0	0	0
0	0	0
0	1	1
0	1	1
0	0	0
0	0	0
0	1	1
0	1	1
0	0	0
0	0	0
0	1	1
0	1	1
1	0	1
1	0	1
1	1	0
1	1	0
1	0	1
1	0	1
1	1	0
1	1	0
1	0	1
1	0	1
1	1	0
1	1	0

2-2) Registre à décalage série : (SISO - SISO2)



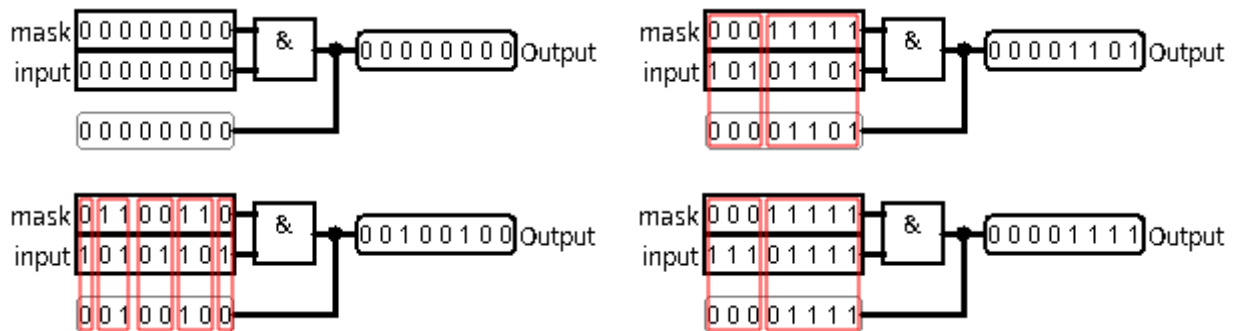


2-3) En totalité : (SRAND4S)

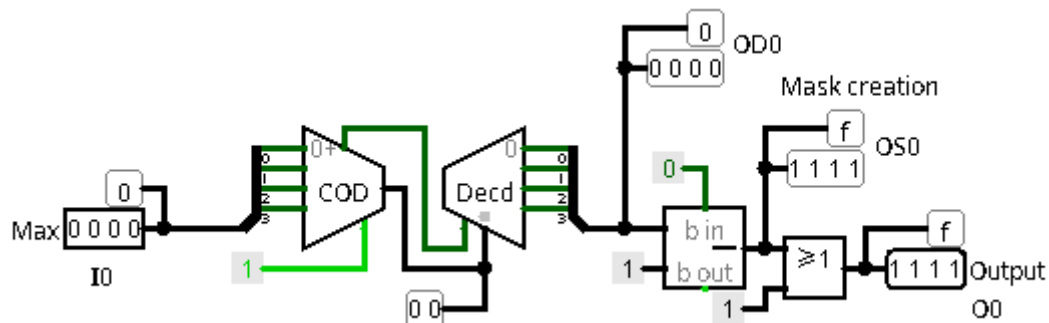


3) Avec valeur maximale :

3-1) C'est quoi un masque : (MASK)

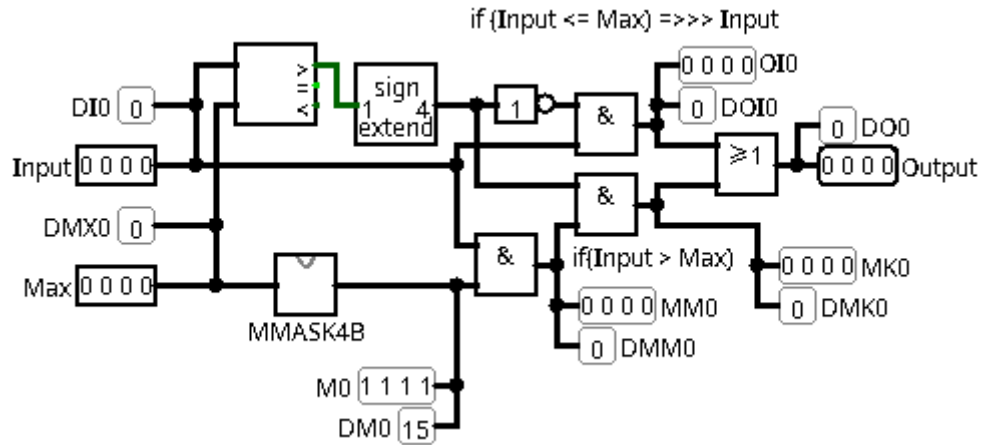


3-2) Création d'un masque pour le nombre aléatoire :



i0					i0V	OD0					OD0V	OS0					OS0V	O0					O0V
0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	15	1	1	1	1	1	15
0	0	0	0	1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1	1
0	0	1	0	0	2	0	0	1	0	0	2	0	0	0	0	1	1	0	0	0	0	1	1
0	0	1	1	0	3	0	0	1	0	0	2	0	0	0	0	1	1	0	0	0	0	1	1
0	1	0	0	0	4	0	1	0	0	0	4	0	0	1	1	1	3	0	0	1	1	1	3
0	1	0	1	0	5	0	1	0	0	0	4	0	0	1	1	1	3	0	0	1	1	1	3
0	1	1	0	0	6	0	1	0	0	0	4	0	0	1	1	1	3	0	0	1	1	1	3
0	1	1	1	0	7	0	1	0	0	0	4	0	0	1	1	1	3	0	0	1	1	1	3
1	0	0	0	0	8	1	0	0	0	0	8	0	1	1	1	1	7	0	1	1	1	1	7
1	0	0	1	0	9	1	0	0	0	0	8	0	1	1	1	1	7	0	1	1	1	1	7
1	0	1	0	0	10	1	0	0	0	0	8	0	1	1	1	1	7	0	1	1	1	1	7
1	0	1	1	0	11	1	0	0	0	0	8	0	1	1	1	1	7	0	1	1	1	1	7
1	1	0	0	0	12	1	0	0	0	0	8	0	1	1	1	1	7	0	1	1	1	1	7
1	1	0	1	0	13	1	0	0	0	0	8	0	1	1	1	1	7	0	1	1	1	1	7
1	1	1	0	0	14	1	0	0	0	0	8	0	1	1	1	1	7	0	1	1	1	1	7
1	1	1	1	0	15	1	0	0	0	0	8	0	1	1	1	1	7	0	1	1	1	1	7

3-3) Valeur max : (RVM4B)



DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
0	0	15	0	0	0	0
0	1	15	1	0	1	1
0	2	15	2	0	2	2
0	3	15	3	0	3	3
0	4	15	4	0	4	4
0	5	15	5	0	5	5
0	6	15	6	0	6	6
0	7	15	7	0	7	7
0	8	15	8	0	8	8
0	9	15	9	0	9	9
0	10	15	10	0	10	10
0	11	15	11	0	11	11
0	12	15	12	0	12	12
0	13	15	13	0	13	13
0	14	15	14	0	14	14
0	15	15	15	0	15	15

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
1	0	1	0	0	0	0
1	1	1	1	0	1	1
1	2	1	0	0	0	0
1	3	1	1	1	0	1
1	4	1	0	0	0	0
1	5	1	1	1	0	1
1	6	1	0	0	0	0
1	7	1	1	1	0	1
1	8	1	0	0	0	0
1	9	1	1	1	0	1
1	10	1	0	0	0	0
1	11	1	1	1	0	1
1	12	1	0	0	0	0
1	13	1	1	1	0	1
1	14	1	0	0	0	0
1	15	1	1	1	0	1

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
2	0	1	0	0	0	0
2	1	1	1	0	1	1
2	2	1	0	0	0	0
2	3	1	1	1	0	1
2	4	1	0	0	0	0
2	5	1	1	1	0	1
2	6	1	0	0	0	0
2	7	1	1	1	0	1
2	8	1	0	0	0	0
2	9	1	1	1	0	1
2	10	1	0	0	0	0
2	11	1	1	1	0	1
2	12	1	0	0	0	0
2	13	1	1	1	0	1
2	14	1	0	0	0	0
2	15	1	1	1	0	1

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
3	0	1	0	0	0	0
3	1	1	1	0	1	1
3	2	1	0	0	0	0
3	3	1	1	1	0	1
3	4	1	0	0	0	0
3	5	1	1	1	0	1
3	6	1	0	0	0	0
3	7	1	1	1	0	1
3	8	1	0	0	0	0
3	9	1	1	1	0	1
3	10	1	0	0	0	0
3	11	1	1	1	0	1
3	12	1	0	0	0	0
3	13	1	1	1	0	1
3	14	1	0	0	0	0
3	15	1	1	1	0	1

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
4	0	3	0	0	0	0
4	1	3	1	0	1	1
4	2	3	2	0	2	2
4	3	3	3	0	3	3
4	4	3	0	0	4	4
4	5	3	1	1	0	1
4	6	3	2	2	0	2
4	7	3	3	3	0	3
4	8	3	0	0	0	0
4	9	3	1	1	0	1
4	10	3	2	2	0	2
4	11	3	3	3	0	3
4	12	3	0	0	0	0
4	13	3	1	1	0	1
4	14	3	2	2	0	2
4	15	3	3	3	0	3

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
5	0	3	0	0	0	0
5	1	3	1	0	1	1
5	2	3	2	0	2	2
5	3	3	3	0	3	3
5	4	3	0	0	4	4
5	5	3	1	1	0	1
5	6	3	2	2	0	2
5	7	3	3	3	0	3
5	8	3	0	0	0	0
5	9	3	1	1	0	1
5	10	3	2	2	0	2
5	11	3	3	3	0	3
5	12	3	0	0	0	0
5	13	3	1	1	0	1
5	14	3	2	2	0	2
5	15	3	3	3	0	3

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
6	0	3	0	0	0	0
6	1	3	1	0	1	1
6	2	3	2	0	2	2
6	3	3	3	0	3	3
6	4	3	0	0	4	4
6	5	3	1	1	0	1
6	6	3	2	2	0	2
6	7	3	3	3	0	3
6	8	3	0	0	0	0
6	9	3	1	1	0	1
6	10	3	2	2	0	2
6	11	3	3	3	0	3
6	12	3	0	0	0	0
6	13	3	1	1	0	1
6	14	3	2	2	0	2
6	15	3	3	3	0	3

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
7	0	3	0	0	0	0
7	1	3	1	0	1	1
7	2	3	2	0	2	2
7	3	3	3	0	3	3
7	4	3	0	0	4	4
7	5	3	1	1	0	1
7	6	3	2	2	0	2
7	7	3	3	3	0	3
7	8	3	0	0	0	0
7	9	3	1	1	0	1
7	10	3	2	2	0	2
7	11	3	3	3	0	3
7	12	3	0	0	0	0
7	13	3	1	1	0	1
7	14	3	2	2	0	2
7	15	3	3	3	0	3

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
8	0	7	0	0	0	0
8	1	7	1	0	1	1
8	2	7	2	0	2	2
8	3	7	3	0	3	3
8	4	7	4	0	4	4
8	5	7	5	0	5	5
8	6	7	6	0	6	6
8	7	7	7	0	7	7
8	8	7	0	0	8	8
8	9	7	1	1	0	1
8	10	7	2	2	0	2
8	11	7	3	3	0	3
8	12	7	4	4	0	4
8	13	7	5	5	0	5
8	14	7	6	6	0	6
8	15	7	7	7	0	7

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
9	0	7	0	0	0	0
9	1	7	1	0	1	1
9	2	7	2	0	2	2
9	3	7	3	0	3	3
9	4	7	4	0	4	4
9	5	7	5	0	5	5
9	6	7	6	0	6	6
9	7	7	7	0	7	7
9	8	7	0	0	8	8
9	9	7	1	1	0	1
9	10	7	2	2	0	2
9	11	7	3	3	0	3
9	12	7	4	4	0	4
9	13	7	5	5	0	5
9	14	7	6	6	0	6
9	15	7	7	7	0	7

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
10	0	7	0	0	0	0
10	1	7	1	0	1	1
10	2	7	2	0	2	2
10	3	7	3	0	3	3
10	4	7	4	0	4	4
10	5	7	5	0	5	5
10	6	7	6	0	6	6
10	7	7	7	0	7	7
10	8	7	0	0	8	8
10	9	7	1	1	0	1
10	10	7	2	2	0	2
10	11	7	3	3	0	3
10	12	7	4	4	0	4
10	13	7	5	5	0	5
10	14	7	6	6	0	6
10	15	7	7	7	0	7

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
11	0	7	0	0	0	0
11	1	7	1	0	1	1
11	2	7	2	0	2	2
11	3	7	3	0	3	3
11	4	7	4	0	4	4
11	5	7	5	0	5	5
11	6	7	6	0	6	6
11	7	7	7	0	7	7
11	8	7	0	0	8	8
11	9	7	1	1	0	1
11	10	7	2	2	0	2
11	11	7	3	3	0	3
11	12	7	4	4	0	4
11	13	7	5	5	0	5
11	14	7	6	6	0	6
11	15	7	7	7	0	7

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
12	0	7	0	0	0	0
12	1	7	1	0	1	1
12	2	7	2	0	2	2
12	3	7	3	0	3	3
12	4	7	4	0	4	4
12	5	7	5	0	5	5
12	6	7	6	0	6	6
12	7	7	7	0	7	7
12	8	7	0	0	8	8
12	9	7	1	1	0	1
12	10	7	2	2	0	2
12	11	7	3	3	0	3
12	12	7	4	4	0	4
12	13	7	5	5	0	5
12	14	7	6	6	0	6
12	15	7	7	7	0	7

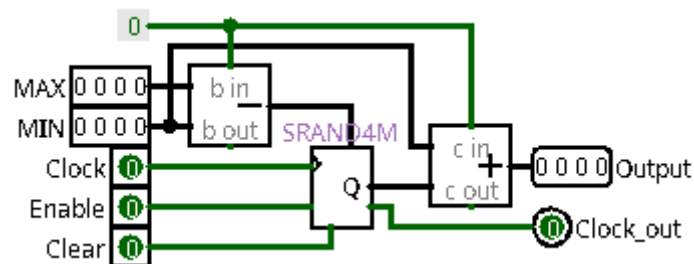
DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
13	0	7	0	0	0	0
13	1	7	1	0	1	1
13	2	7	2	0	2	2
13	3	7	3	0	3	3
13	4	7	4	0	4	4
13	5	7	5	0	5	5
13	6	7	6	0	6	6
13	7	7	7	0	7	7
13	8	7	0	0	8	8
13	9	7	1	1	0	1
13	10	7	2	2	0	2
13	11	7	3	3	0	3
13	12	7	4	4	0	4
13	13	7	5	5	0	5
13	14	7	6	6	0	6
13	15	7	7	7	0	7

DMX0	DIO	DM0	DMM0	DMK0	DOIO	DOO
14	0	7	0	0	0	0
14	1	7	1	0	1	1
14	2	7	2	0	2	2
14	3	7	3	0	3	3
14	4	7	4	0	4	4
14	5	7	5	0	5	5

DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0
0	0	0	0	1	0	0	2	0	0	3	0	0	4	0	0	5	0	0	6	0	0	7	0
1	0	1	1	1	1	1	2	1	1	3	1	1	4	1	1	5	1	1	6	1	1	7	1
2	0	2	2	1	0	2	2	2	2	3	2	2	4	2	2	5	2	2	6	2	2	7	2
3	0	3	3	1	1	3	2	1	3	3	3	3	4	3	3	5	3	3	6	3	3	7	3
4	0	4	4	1	0	4	2	0	4	3	0	4	4	4	4	5	4	4	6	4	4	7	4
5	0	5	5	1	1	5	2	1	5	3	1	5	4	1	5	5	5	5	6	5	5	7	5
6	0	6	6	1	0	6	2	0	6	3	0	6	4	2	6	5	2	6	6	6	6	7	6
7	0	7	7	1	1	7	2	1	7	3	1	7	4	3	7	5	3	7	6	3	7	7	7
8	0	8	8	1	0	8	2	0	8	3	0	8	4	0	8	5	0	8	6	0	8	7	0
9	0	9	9	1	1	9	2	1	9	3	1	9	4	1	9	5	1	9	6	1	9	7	1
10	0	10	10	1	0	10	2	0	10	3	0	10	4	2	10	5	2	10	6	2	10	7	2
11	0	11	11	1	1	11	2	1	11	3	1	11	4	3	11	5	3	11	6	3	11	7	3
12	0	12	12	1	0	12	2	0	12	3	0	12	4	0	12	5	0	12	6	0	12	7	0
13	0	13	13	1	1	13	2	1	13	3	1	13	4	1	13	5	1	13	6	1	13	7	1
14	0	14	14	1	0	14	2	0	14	3	0	14	4	2	14	5	2	14	6	2	14	7	2
15	0	15	15	1	1	15	2	1	15	3	1	15	4	3	15	5	3	15	6	3	15	7	3

DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0	DIO	DMX0	DO0
0	8	0	0	9	0	0	10	0	0	11	0	0	12	0	0	13	0	0	14	0	0	15	0
1	8	1	1	9	1	1	10	1	1	11	1	1	12	1	1	13	1	1	14	1	1	15	1
2	8	2	2	9	2	2	10	2	2	11	2	2	12	2	2	13	2	2	14	2	2	15	2
3	8	3	3	9	3	3	10	3	3	11	3	3	12	3	3	13	3	3	14	3	3	15	3
4	8	4	4	9	4	4	10	4	4	11	4	4	12	4	4	13	4	4	14	4	4	15	4
5	8	5	5	9	5	5	10	5	5	11	5	5	12	5	5	13	5	5	14	5	5	15	5
6	8	6	6	9	6	6	10	6	6	11	6	6	12	6	6	13	6	6	14	6	6	15	6
7	8	7	7	9	7	7	10	7	7	11	7	7	12	7	7	13	7	7	14	7	7	15	7
8	8	8	8	9	8	8	10	8	8	11	8	8	12	8	8	13	8	8	14	8	8	15	8
9	8	1	9	9	9	9	10	9	9	11	9	9	12	9	9	13	9	9	14	9	9	15	9
10	8	2	10	9	2	10	10	10	10	11	10	10	12	10	10	13	10	10	14	10	10	15	10
11	8	3	11	9	3	11	10	3	11	11	11	11	12	11	11	13	11	11	14	11	11	15	11
12	8	4	12	9	4	12	10	4	12	11	4	12	12	12	12	13	12	12	14	12	12	15	12
13	8	5	13	9	5	13	10	5	13	11	5	13	12	5	13	13	13	13	14	13	13	15	13
14	8	6	14	9	6	14	10	6	14	11	6	14	12	6	14	13	6	14	14	14	14	15	14
15	8	7	15	9	7	15	10	7	15	11	7	15	12	7	15	13	7	15	14	7	15	15	15

4) Avec valeur minimale :



5) 8 bits :

